Input data to METIS read from IMAS IDSs: METIS configuration and external data

Introduction

METIS is interfaced with IMAS through the function “metis4imas”. METIS can read IDSs (IMAS Interface Data Structures) as input and/or write results in IDSs, depending of the parametrization of this function. The use of this function is detailed in the help of the function (command “help metis4imas” in MATLAB console). The detail of the data output in IDSs by METIS is available in the file “metis_imas_output_documentation.md” in the directory “doc” from the root of the METIS installation folder.

METIS input is separated in two set of data:

- References and parameters that configure the code and are stored in the MATLAB data structure “z0dinput” (see “Howto_METIS_final.pdf” and “metis_documentation_for_input_data_structure.pdf” in ”doc” directory for details).

- External data that allow to substitute quantities computed by METIS with prescribed quantities. The details are described in the mfile “external_data_rule_for_METIS.m”, with sample MATLAB code for the implementation. External data are automatically used in METIS when they are available. To finish using a external data, it must be deleted from the Matlab root appdata workspace..

The present document lists the data used as input to METIS from IDS data structures. It is split in two parts. The first one describes the data needed to configure METIS and the second part describes the optional data used as external data in METIS. Some external METIS data can’t be mapped on any existing IMAS data and will not be available from the IMAS METIS interface.

Data for METIS configuration

METIS configuration is read from the “pulse_schedule” IDS. The references (scalar time dependent quantities) are read from the main pulse_schedule data structure. The METIS parameters stored in z0dinput.option substructure (see “metis_documentation_for_parameters_expert_mode.pdf” for details) is read from pulse_schedule.code.parameters field and are encoded in an XML formatted string. An example of such an XML string is available in the file “metis4imas.xml” in the root directory of the METIS installation.

The fields used for the references are described above. The first list contains the mandatory ones and the second the optional onestaken into account only when the corresponding field is filled. Some references point to more than one IMAS field name as the name may have evolved between different versions of the IMAS dictionary: please check the relevant name for your IMAS version in IMAS documentation. Units are indicated in IMAS documentation.

METIS supports data stored both directly in “reference” or in “reference.data” (obsolete since DD version 3.40.0). Only the former structure in mentioned in the next paragraphs.

The input time vector is the one stored in “pulse_schedule.time”.

Mandatory data

1. Plasma current (z0dinput.cons.ip)
   pulse_schedule.flux_control.i_plasma.reference
   pulse_schedule.flux_control.ip.reference

2. Line averaged density (real(z0dinput.cons.nbar))
   pulse_schedule.density_control.n_e_line.reference.data.reference

   remark: only the definition corresponding to
   pulse_schedule.density_control.n_e_line.n_e_line_method.index=3 is supported (integral of a 1D core profile over rho_tor_norm up to the LCFS).

3. Zeff (z0dinput.cons.zeff)
   pulse_schedule.density_control.zeff.reference

   remark: only the definition corresponding to
   pulse_schedule.density_control.zeff.zeff_method.index=3 is supported (integral of a 1D core profile over rho_tor_norm up to the LCFS).

4. Major radius (z0dinput.geo.R)
   pulse_schedule.position_control.geometric_axis.r.reference

   if not available, a possible backup solution is:
   pulse_schedule.position_control.magnetic_axis.r.reference

5. Vertical shift (z0dinput.geo.z0)
   pulse_schedule.position_control.geometric_axis.z.reference

   if not available, a possible backup solution is:
   pulse_schedule.position_control.magnetic_axis.z.reference

6. Minor radius (z0dinput.geo.a)
   pulse_schedule.position_control.minor_radius.reference

7. Vacuum toroidal magnetic field at position of major radius z0dinput.geo.R (z0dinput.geo.b0)
   pulse_schedule.tf.b_field_tor_vacuum_r.reference

Optional data

1. Loop voltage (z0dinput.cons.flux)
   pulse_schedule.flux_control.loop_voltage.reference
   pulse_schedule.flux_control.v_loop.reference

   remark: the IMAS loop voltage data is integrated over time to get the flux and an offset is computed to match the initial flux set in METIS at the start of the METIS simulation.

2. Confinement enhancement factor (z0dinput.cons.hmore)
   pulse_schedule.flux_control.beta_normal.reference
   pulse_schedule.flux_control.beta_tor_norm.reference

   used simultaneously with

   pulse_schedule.flux_control.li_3.reference

   remark: the rule used to link “hmore” and beta normalized is:
   hmore = beta_normal / (4 * l_i)
   The reference l_i should also be filled in this case.

3. Isotope ratio (z0dinput.cons.iso, real part for main species and imaginary part for minority fuel)
   - for D-T plasma: pulse_schedule.density_control.n_t_over_n_d.reference

   - for D-He3 (T) plasma: pulse_schedule.density_control.n_he3_over_n_d.reference
     and pulse_schedule.density_control.n_t_over_n_d.reference
     if available , otherwise pulse_schedule.density_control.n_t_over_n_d.reference

   - for H-B11 plasma: pulse_schedule.density_control.n_b_over_n_h.reference
     if available, otherwise pulse_schedule.density_control.n_t_over_n_d.reference

4. Gas puff (imag(z0dinput.cons.nbar))
   This is the sum of the flux of each valve times the charge of species injected. If there is more than one element, the charge is computed by summing on all the element weighted by the fraction of each element.

   pulse_schedule.density_control.valve{k}.flow_rate.reference

   and if it is not a simple element:
   pulse_schedule.density_control.valve{k}.species{l}.element{m}.z_n
   and
   pulse_schedule.density_control.valve{k}.species{l}.fraction

5. ICRH power (z0dinput.cons.picrh)
   pulse_schedule.ic.launcher{1}.power.reference
   pulse_schedule.ic.antenna{1}.power.reference

pulse_schedule.ic.power.refrence

6. LHCD power (z0dinput.cons.plh)
   pulse_schedule.lh.antenna{1}.power.reference
   pulse_schedule.lh.launcher{1}.power.reference
   pulse_schedule.lh.power.refrence

7. NBI power (z0dinput.cons.pnbi, only one box)
   sum_k(pulse_schedule.nbi.unit{1}.power.reference)
   pulse_schedule.nbi.power.reference

8. ECRH power (z0dinput.cons.pecrh)
   pulse_schedule.ec.beam{1}.power_launched.reference
   pulse_schedule.ec.antenna{1}.power.reference
   pulse_schedule.ec.power_launched.reference

9. Second ECRH power (z0dinput.cons.plh)
   pulse_schedule.ec.beam{2}.power_launched.reference
   pulse_schedule.ec.antenna{2}.power.reference

   remark: In this case, internally METIS uses the LH slot with a specific parametrization. LHCD can't be used when a second ECRH source is declared.

10. ECRH maximum power deposition positition (z0dinput.cons.xece)
    pulse_schedule.ec.beam{1}.deposition_rho_tor_norm.reference
    pulse_schedule.ec.antenna{1}.deposition_rho_tor_norm.reference.data

    remark: the internal radial coordinate in METIS is the Lao coordinate x=r(rho)/aminor. As there is no available information to perform a coordinate change at this point, "deposition_rho_tor_norm" is used as a proxy of x.

11. Plasma elongation (z0dinput.geo.K)
    pulse_schedule.position_control.elongation_lower.reference.data

    if not available, a possible backup solution is:
    (pulse_schedule.position_control.elongation_upper.reference +
    pulse_schedule.position_control.elongation_lowr.reference)/2

12. Plasma triangularity (z0dinput.geo.d)
    pulse_schedule.position_control.triangularity.reference

    if not available, a possible backup solution is:
    (pulse_schedule.position_control.triangulaity_upper.reference +
    pulse_schedule.position_control.triangularity_lowr.reference)/2

13. LCFS provided as a set of (R,Z) points
    pulse_schedule.position_control.boundary_outline{k}.r.reference
    And
    pulse_schedule.position_control.boundary_outline{k}.z.reference

    remark: if the LCFS is provided as a set of (R,Z) points, its geometrical moments (zd0input.geo) are recomputed from these data.

METIS external data from IDSs

This section describes the IDSs fields needed to activate each external data for METIS including time synchronization, coordinate change and COCOS management.

Time synchronization

A synchronization of the clock between IMAS and METIS is needed. The clock for external data is taken from IDS data. If the “time” field at the root of the IDS data structure is available, this data will be used, otherwise the “time” field at deeper levels (i.e. profiles_1d.time, time_slice.time , …) is used.

For the MUSCLE3 METIS actor, an offset defined in actor setting “timeoffset” is subtracted from the IMAS external data time. In the METIS GUI, an offset can be defined when external data from IMAS are read.

Coordinates mapping

The internal radial coordinate in METIS is the Lao coordinate x = r(rho)/aminor. A coordinate change is performed between IMAS rho_tor_norm that should be defined in used IDSs and the Lao coordinate. The coordinate change requires the following data from IDS equilibrium:

- equi_ids.time_slice{:}.profiles_1d.rho_tor_norm
- equi_ids.time_slice{:}.profiles_1d.r_outboard
- equi_ids.time_slice{:}.profiles_1d.r_inboard

If the IDS equilibrium is not available, a backup solution will be used using either:

- Previous computed METIS time slice in evolution mode and in METIS MUSCLE 3 actor
- Load METIS simulation in Matlab workspace for standard METIS run and METIS data from METIS file declared in METIS MUSCLE 3 actor

As a last backup solution, rho_tor_norm will be interpreted as a proxy of x.

COCOS handling

COCOS for input data is assumed to be 11 for IMAS dictionary 3.x and 17 for IMAS dictionary 4.x. The signs of the plasma current and of the toroidal magnetic field are deduced from:

- global_quantities.ip
- vacuum_toroidal_field.b0

from either the equilibrium, plasma_profiles or core_profiles IDS.

If inconsistences are detected or when these two fields are not available, the internal METIS setting for COCOS and signs are used.

Selection of the source

In plasma_sources or core_sources IDS, the index for the corresponding source is selected using the “identifier.name” content.

Available external data from IMAS IDSs

Next paragraph lists the IMAS data needed for each external data feature.

equilibrium

Vacuum toroidal magnetic field

equilibrium.vacuum_toroidal_field.b0

equilibrium.vacuum_toroidal_field.r0

Grid quantities

equilibrium.time_slice{:}.profiles_1d.psi

equilibrium.time_slice{:}.profiles_1d.phi

equilibrium.time_slice{:}.profiles_1d.rho_tor

equilibrium.time_slice{:}.profiles_1d.r_outboard

equilibrium.time_slice{:}.profiles_1d.r_inboard

Global quantities

equilibrium.time_slice{:}.global_quantities.ip

equilibrium.time_slice{:}.global_quantities.li_3

equilibrium.time_slice{:}.global_quantities.beta_pol

equilibrium.time_slice{:}.global_quantities.q_95

equilibrium.time_slice{:}.global_quantities.q_axis

Metric quantities

equilibrium.time_slice{:}.profiles_1d.gm1

equilibrium.time_slice{:}.profiles_1d.gm2

equilibrium.time_slice{:}.profiles_1d.gm3

equilibrium.time_slice{:}.profiles_1d.gm7

equilibrium.time_slice{:}.profiles_1d.gm8

equilibrium.time_slice{:}.profiles_1d.gm9

equilibrium.time_slice{:}.profiles_1d.dpsi_drho_tor

equilibrium.time_slice{:}.profiles_1d.darea_drho_tor

equilibrium.time_slice{:}.profiles_1d.dvolume_drho_tor

Geometrical quantities

equilibrium.time_slice{:}.profiles_1d.volume

equilibrium.time_slice{:}.profiles_1d.area

equilibrium.time_slice{:}.profiles_1d.surface

equilibrium.time_slice{:}.profiles_1d.geometric_axis.r

equilibrium.time_slice{:}.profiles_1d.geometric_axis.z

equilibrium.time_slice{:}.profiles_1d.elongation

equilibrium.time_slice{:}.profiles_1d.triangularite_upper

equilibrium.time_slice{:}.profiles_1d. triangularite_lower

equilibrium.time_slice{:}.boundary.minor_radius

equilibrium.time_slice{:}.boundary.geometric_axis.r

equilibrium.time_slice{:}.boundary.geometric_axis.z

equilibrium.time_slice{:}.boundary.elongation

equilibrium.time_slice{:}.boundary.triangularity

equilibrium.time_slice{:}.boundary.x_point{:}.r

equilibrium.time_slice{:}.boundary.x_point{:}.z

equilibrium.time_slice{:}.boundary.outline.r

equilibrium.time_slice{:}.boundary.outline.z

equilibrium.time_slice{:}.boundary.trapped_fraction

remark: if the field trapped_fraction is empty, the internal METIS computation is used as a backup.

Physical quantities

equilibrium.time_slice{:}.profiles_1d.j_tor or equilibrium.time_slice{:}.profiles_1d.j_phi

equilibrium.time_slice{:}.profiles_1d.j_parallel

equilibrium.time_slice{:}.profiles_1d.q

equilibrium.time_slice{:}.profiles_1d.pressure

equilibrium.time_slice{:}.profiles_1d.f

current diffusion

The current diffusion solved externally to METIS needs the same set of data than the external equilibrium. It can be activated independently of the external equilibrium but the external equilibrium must be used when the current diffusion is solved externally.

electron temperature profile

plasma_profiles.profiles_1d{:}.electrons.temperature

or

core_profiles.profiles_1d{:}.electrons.temperature

electron density profile

plasma_profiles.profiles_1d{: }.electrons.density

or

core_profiles.profiles_1d{: }.electrons.density

remark: the profile shape is read from IDS data but the final value is renormalized to match the nbar reference of METIS.

ion temperature profile

plasma_profiles.profiles_1d{:}.t_i_average

or

core_profiles.profiles_1d{:}.t_i_average

Effective charge profile

plasma_profiles.profiles_1d{:}.zeff

or

core_profiles.profiles_1d{:}.zeff

remark: the profile shape is read from IDS data but the final value is renormalized to match the Zeff reference of METIS.

LHCD source: ('LH','LHCD')

plasma_sources.source{'LH'}.profiles_1d{ :}.j_parallel

plasma_source.source{'LH'}.profiles_1d{ :}.electrons.energy

or

core_sources.source{'LH'}.profiles_1d{ :}.j_parallel

core_source.source{'LH'}.profiles_1d{ :}.electrons.energy

remark : these data are renormalized to the LH METIS reference using METIS internal volume element and preserving the current drive efficiency.

ECRH/ECCD source: ('EC','ECRH','ECCD')

plasma_sources.source{'EC'}.profiles_1d{ :}.j_parallel

plasma_source.source{'EC'}.profiles_1d{ :}.electrons.energy

or

core_sources.source{‘EC’}.profiles_1d{ :}.j_parallel

core_source.source{‘EC’}.profiles_1d{ :}.electrons.energy

remark : these data are renormalized to the ECRH METIS reference using METIS internal volume element and preserving the current drive efficiency.

ICRH/FWEH source: ('IC','ICRH','FW','FWCD')

plasma_sources.source{*}.profiles_1d{ :}.j_parallel

plasma_sources.source{*}.profiles_1d{ :}.electrons.energy

plasma_sources.source{*}.profiles_1d{ :}.total_ion_energy

or

core_sources.source{*}.profiles_1d{ :}.j_parallel

core_sources.source{*}.profiles_1d{ :}.electrons.energy

core_sources.source{*}.profiles_1d{ :}.total_ion_energy

remark : these data are renormalized to the ICRH METIS reference using METIS internal volume element and preserving the current drive efficiency and balance between electron and ion heating.

NBI source: ('NBI','NBICD')

plasma_sources.source{‘NBI’}.profiles_1d{ :}.j_parallel

plasma_sources.source{‘NBI’}.profiles_1d{ :}.electrons.energy

plasma_sources.source{‘NBI’}.profiles_1d{ :}.total_ion_energy

or

core_sources.source{‘NBI’}.profiles_1d{ :}.j_parallel

core_sources.source{‘NBI’}.profiles_1d{ :}.electrons.energy

core_sources.source{‘NBI’}.profiles_1d{ :}.total_ion_energy

remark : these data are renormalized to the NBI METIS reference using METIS internal volume element and preserving the current drive efficiency and balance between electron and ion heating.

Line radiative power density source ('LINE','LINERADIATION','LINE_RADIATION','PLINE','PRAD')

plasma_sources.source{*}.profiles_1d{ :}.electrons.energy

or

core_sources.source{*}.profiles_1d{ :}.electrons.energy

Runaway electrons ('RUNAWAY','RUNAWAYS')

plasma_sources.source{*}.profiles_1d{ :}.j_parallel

or

core_sources.source{*}.profiles_1d{ :}.j_parallel

and

plasma_profiles.profiles_1d{:}.j_ohmic

or

core_profiles.profiles_1d{:}.j_ohmic

and

equilibrium.time_slice{:}.profiles_1d.rho_tor

equilibrium.time_slice{:}.profiles_1d.darea_drho_tor

Toroidal momentum profile

Once of the following fields:

- plasma_profiles.profiles_1d{:}.momentum_tor
- plasma_profiles.profiles_1d{:}.momentum_phi
- core_profiles.profiles_1d{:}.momentum_tor
- core_profiles.profiles_1d{ :}.momentum_phi

and

plasma_profiles.profiles_1d{:}.electrons.density

or

core_profiles.profiles_1d{:}.electrons.density

and

equilibrium.time_slice{:}.profiles_1d.gm1